

APO-PROPRANOLOL

Propranolol Tablets HCl 10mg and 40mg Beta-Adrenergic Receptor Blocking Agent

Pharmacology:

Propranolol is a beta-adrenergic receptor blocking agent. When access to beta-adrenergic receptor sites is blocked by propranolol, the chronotropic, inotropic, and vasodilator responses to beta-adrenergic stimulation are decreased proportionately. Beta-adrenergic blockade is useful in some clinical conditions in which sympathetic activity is excessive or inappropriate, and, therefore, detrimental to the patient. Sympathetic stimulation is however, vital in some situations, (e.g., in patients with AV block or with a severely damaged heart) and should be preserved. The basic objective of beta-adrenergic blockade is to decrease adverse sympathetic stimulation but not to the degree that impairs necessary sympathetic support. Beta-blockade results in bronchial constriction by interfering with endogenously or exogenously induced bronchodilation (see Contraindications and Warnings). The mechanism of the antihypertensive effects of propranolol has not been established. Among the factors that may be involved are: decreased cardiac output; inhibition of renin release by the kidneys; and, diminution of tonic sympathetic nerve outflow from vasomotor centres in the brain. It has been suggested, but not established, that propranolol may achieve a better antihypertensive effect in patients with normal or elevated plasma renin activity than those with low PRA. Propranolol may reduce the oxygen requirement of the heart at any level of effort by blocking catecholamine-induced increases in the heart rate, systolic blood pressure, and the velocity and extent of myocardial contraction. On the other hand, the drug may increase oxygen requirements by increasing left ventricular fiber length, and diastolic pressure, and systolic ejection period. When the net effect is beneficial in anginal patients, it manifests during exercise or stress by delaying the onset of pain and reducing the incidence and severity of anginal attacks. Propranolol exerts antiarrhythmic effects in concentrations producing beta-adrenergic blockade, which appears to be its antiarrhythmic mechanism of action. Beta-adrenergic blockade is of importance in the management of arrhythmias caused by increased levels of circulating catecholamines or enhanced sensitivity of the heart to catecholamines (arrhythmias associated with pheochromocytoma, thyrotoxicosis, exercise). Propranolol is almost completely absorbed from the gastrointestinal tract and undergoes extensive presystemic (or "first pass") elimination due to its high hepatic clearance. Inter-individual variations in circulating drug concentrations due to this "first pass effect" have been documented and differ according to a number of factors, including genetic make-up. Peak plasma concentrations of propranolol are attained in 60 to 90 minutes. The plasma half-life is 2 to 3 hours, whereas the duration of pharmacological effect is longer.

Indications:

Hypertension: For the treatment of mild to moderate hypertension. It is usually used in combination with other drugs, particularly a thiazide diuretic. Propranolol may, however, in certain patients be used alone or as an initial agent for those patients for whom, in the judgment of the physician, treatment should be started with a beta-blocker rather than a diuretic. The combination of propranolol with thiazide-like diuretics and/or peripheral vasodilators has been shown to be compatible and generally more effective than using propranolol alone. Experience with most commonly used antihypertensive agents has not suggested evidence of incompatibility. Propranolol is not recommended for the emergency treatment of hypertensive crises. Angina Pectoris: For the prophylaxis of angina pectoris. Cardiac Arrhythmias: Supraventricular arrhythmias: paroxysmal atrial tachycardias, particular those arrhythmias induced by catecholamines or digitalis or associated with Wolff-Parkinson-White syndrome (see WPW under Warnings); persistent sinus tachycardia which is non-compensatory and impairs the well-being of the patient; tachycardias and arrhythmias due to thyrotoxicosis when causing distress or increased hazard and when immediate effect is necessary as adjunctive, short-term (2 to 4 weeks) therapy (may be used with, but not in place of, specific therapy) (see Thyrotoxicosis under Warnings); persistent atrial extrasystoles which impair the well-being of the patient and do not respond to conventional measures; and atrial flutter and fibrillation when ventricular rate cannot be controlled by digitalis alone, or when digitalis is contraindicated. Tachyarrhythmias of Digitalis Intoxication: If digitalis-induced tachyarrhythmias persist following discontinuance of digitalis and correction of electrolyte abnormalities, they may be reversible with propranolol. Severe bradycardia may occur (see Overdose). Migraine: For the prophylaxis of migraine headache. It is not indicated for the treatment of acute migraine attacks. Hypertrophic Subaortic Stenosis: In the management of hypertrophic subaortic stenosis, especially for treatment of exertional or other stress-induced angina, palpitations, and syncope. Propranolol may also improve exercise performance. The effectiveness of propranolol in this disease is apparently due to a reduction of the elevated outflow pressure gradient exacerbated

by beta-adrenergic receptor stimulation. Clinical improvement may be temporary. Pheochromocytoma: After primary treatment with an alpha-adrenergic blocking agent has been instituted, propranolol may be useful as adjunctive therapy if the control of tachycardia becomes necessary before or during surgery. It is hazardous to use propranolol unless alpha-adrenergic blocking drugs are already in use since this would predispose to serious blood pressure rise. Blocking only the peripheral dilator (beta) action of epinephrine leaves its constrictor (alpha) action unopposed. In the event of hemorrhage or shock, producing both beta and alpha-blockade is contraindicated since the combination prevents the increase in heart rate and peripheral vasoconstriction needed to maintain blood pressure. In inoperable or metastatic pheochromocytoma, propranolol may be useful as an adjunct to the management of symptoms due to excessive beta-adrenergic receptor stimulation.

Contraindications:

Bronchospasm, including bronchial asthma; allergic rhinitis during pollen season; sinus bradycardia and greater-than-first-degree block; cardiogenic shock; right ventricular failure secondary to pulmonary hypertension; congestive heart failure (see Warnings) unless the failure is secondary to a tachyarrhythmia treatable with propranolol.

Warnings:

Cardiac Failure: Sympathetic stimulation is a vital component supporting circulatory function in congestive heart failure; therefore, inhibition by means of beta-adrenergic blockade is a potential hazard as it may further depress myocardial contractility and precipitate cardiac failure. Propranolol acts selectively without abolishing the inotropic action of digitalis on the heart muscle (i.e. that of supporting the strength of myocardial contractions). In patients already receiving digitalis, the positive inotropic action of digitalis may be reduced by propranolol's negative inotropic effect. The effects of propranolol and digitalis are additive in depressing AV conduction. Patients with No History of Cardiac Failure: Continued depression of the myocardium over a period of time can, in some patients, lead to cardiac failure. In rare instances, this has been observed during propranolol therapy. Therefore, at the first sign or symptom of impending cardiac failure, fully digitalize patients and/or give a diuretic. Observe the response closely. If cardiac failure continues, despite adequate digitalization and diuretic therapy, withdraw propranolol immediately. If tachyarrhythmia is being controlled, maintain the patient on combined therapy. Follow closely until the threat of cardiac failure is over. Abrupt Cessation of Therapy in Angina Pectoris: Severe exacerbation of angina and the occurrence of myocardial infarction have been reported in some patients with angina pectoris following abrupt discontinuation of propranolol therapy. Therefore, when discontinuation is planned in patients with angina pectoris gradually reduce the dosage over a period of about 2 weeks and observe the patient carefully. Maintain the same frequency of administration. In situations of greater urgency, reduce dosage stepwise, in 4 days, under close observation. If angina markedly worsens or acute coronary insufficiency develops, reinstitute treatment with propranolol promptly, at least temporarily. In addition, warn patients with angina pectoris against abrupt discontinuation of propranolol. Oculomucocutaneous Syndrome: Various skin rashes and conjunctival xerosis have been reported in patients treated with beta-blockers, including propranolol. A severe oculomucocutaneous syndrome, whose signs include conjunctivitis sicca and psoriasiform rashes, otitis, and sclerosing serositis, has occurred with the long-term use of one beta-adrenergic blocking agent. This syndrome has not been observed with propranolol; however, physicians should be alert to the possibility of such reactions and discontinue treatment if they occur. Patients with Thyrotoxicosis: Possible deleterious effects from long-term use have not yet been adequately appraised. Special consideration should be given to propranolol's potential for aggravating congestive heart failure. It may mask the clinical signs of developing or continuing hyperthyroidism or its complications, and may give a false impression of improvement. Therefore, abrupt withdrawal of propranolol may be followed by an exacerbation of symptoms of hyperthyroidism, including thyroid storm. This may be another instance where the drug should be withdrawn slowly by reducing dosage. Propranolol HCl does not distort thyroid function tests. Patients with Wolff-Parkinson-White Syndrome: Use with caution. Several cases have been reported in which, after propranolol treatment, the tachycardia was replaced by a severe bradycardia requiring a demand pacemaker. In one patient, this occurred after an initial dose of 5 mg of propranolol. Patients Undergoing Elective or Emergency Surgery: The management of patients with angina, being treated with beta-blockers and undergoing elective or emergency surgery, is controversial because beta-adrenergic receptor blockade impairs the ability of the heart to respond to beta-adrenergically mediated reflex stimuli. However, abrupt discontinuation of propranolol therapy may be followed by severe complications (see Warnings). Some patients receiving beta-adrenergic blocking agents have been subject to protracted severe hypotension during anesthesia. Difficulty in restarting and maintaining the heartbeat has also been reported. For these reasons, in patients with angina undergoing elective surgery,

withdraw propranolol therapy gradually following the recommendation given under Abrupt Cessation of Therapy (see Warnings). According to available evidence, all clinical and physiologic effects of beta-blockade are no longer present 48 hours after cessation of medication. In emergency surgery, since propranolol is a competitive inhibitor of beta-adrenergic receptor agonists, its effects may be reversed, if necessary, by sufficient doses of such agonists as isoproterenol or levaterenol. Avoid using anesthesia with agents which maintain cardiac contractility by virtue of their effect on catecholamine release (e.g., ether). Patients Prone to Nonallergic Bronchospasm, e.g., chronic bronchitis, emphysema, bronchiectasis: Administer propranolol with caution since it may block bronchodilation produced by endogenous and exogenous catecholamine stimulation of beta-adrenergic receptors. Patients with Diabetes and those Subject to Hypoglycemia: Propranolol may block premonitory signs and symptoms (pulse rate and pressure changes) of acute hypoglycemia. This is especially important to keep in mind in patients with labile diabetes. Hypoglycemic attacks may be accompanied by a precipitous elevation of blood pressure. Pregnancy: Safe use has not been established. Use of any drug in pregnancy or in women of childbearing potential requires that the possible risk to mother and/or fetus be weighed against the expected therapeutic benefit. Perinatal complications, such as small placenta and intrauterine growth retardation, have been reported in a few cases where the mother took propranolol during pregnancy. Some infants born to mothers treated with propranolol were reported to have hypoglycemia and/or bradycardia. Children: While experience with children under 12 is limited, the indications for which propranolol would be recommended occur infrequently in childhood. Although reports fail to indicate that children respond in any manner different than that of the adult, physicians are advised to undertake treatment with caution. Epinephrine and beta-blockers: There may be increased difficulty in treating an allergic type reaction in patients on beta-blockers. In these patients, the reaction may be more severe due to pharmacologic effects of the beta-blockers and problems with fluid changes. Epinephrine should be administered with caution since it may not have its usual effects in the treatment of anaphylaxis. On the one hand, larger doses of epinephrine may be needed to overcome the bronchospasm, while on the other hand, these doses can be associated with excessive alpha adrenergic stimulation with consequent hypertension, reflex bradycardia and heart block and possible potentiation of bronchospasm. Alternatives to the use of large doses of epinephrine include vigorous supportive care such as fluids and the use of beta agonists including parenteral salbutamol or isoproterenol to overcome bronchospasm and norepinephrine to overcome hypotension.

Precautions:

Some slowing of heart due to unopposed vagal activity is usual in patients receiving propranolol. However, occasionally severe bradycardia occurs and leads to vertigo, syncopal attacks, or orthostatic hypotension. Patients, especially those with limited cardiac reserve, should be monitored for signs of excessive bradycardia. Should the patient become symptomatic, decrease propranolol dosage or, if necessary, discontinue. If it is essential to correct the bradycardia, i.v. atropine or isoproterenol should be considered. It has been reported that administration of propranolol to control cardiac arrhythmias in acute myocardial infarction has caused marked reduction in cardiac output. Therefore, keep doses to a minimum in patients with severe myocardial infarction. Use the drug with caution in patients with impaired renal and hepatic function.

Drug Interactions :

Prior administration of other antiarrhythmic, cardiac-depressant drugs, such as procainamide or quinidine, may potentiate propranolol's cardiac-depressant activities. Prior digitalization may be indicated; atropine should be at hand to control bradycardia.

Bradycardia, cardiac arrest, and ventricular fibrillation have been reported shortly after initiation of beta-blocker treatment in patients receiving amiodarone.

Significant increases in plasma-lignocaine concentrations have occurred during concomitant therapy with propranolol owing to a reduction in the clearance of lignocaine from plasma. Concomitant administration of propafenone and propranolol resulted in increases in serum-propranolol concentrations but only modest enhancement of beta-blocking activity.

Concurrent administration of calcium-channel blockers and beta-blockers has resulted in hypotension, bradycardia, conduction defect, and cardiac failure. Beta-blockers should be avoided in combination with cardiodepressant calcium-channel blockers such as verapamil and diltiazem. Although they are reportedly safe in combination with the dihydropyridines such as nifedipine, heart failure and severe hypotension have been reported. Reported pharmacokinetic interactions include increased plasma concentrations of propranolol with concurrent administration of diltiazem or verapamil, and increased plasma concentrations of propranolol with nifedipine. The combination of propranolol with a thiazide-like diuretic and/or peripheral vasodilator produces a

great fall in blood pressure than either drug alone. This occurs regardless of which drug is administered first. The same degree of blood pressure control can be achieved by lower than usual dosages of each drug. Therefore, when using such combined therapy, monitor the dosages carefully until the patient is stabilized. Observe patients receiving catecholamine-depleting drugs, such as reserpine or guanethidine, closely if propranolol is administered concomitantly. The added catecholamine-blocking action of this drug may produce an excessive reduction of the resting sympathetic nervous activity. In patients on long term treatment with propranolol, make laboratory determinations at regular intervals.

Plasma concentrations of propranolol may be reduced by concomitant rifampicin administration.

A report of increased plasma-propranolol concentrations and bioavailability and reduced metabolism in healthy subjects also given chlorpromazine.

Adverse Effects:

The most serious adverse effects that may be encountered with propranolol are congestive heart failure and bronchospasm (see Contraindications, Warnings, and Precautions). Gastrointestinal disturbances (anorexia, nausea, vomiting, diarrhea, abdominal pain) are the most common adverse effects reported. Other less frequently reported adverse effects (in descending order) are: cold extremities and exacerbation of Raynaud's phenomenon; congestive heart failure; sleep disturbances including vivid dreams; dizziness; fatigue; and, bronchospasm. Cardiovascular: congestive heart failure (see Warnings). Secondary effects of decreased cardiac output may include: syncope, vertigo, lightheadedness, decreased renal perfusion and, rarely, postural hypotension; intensification of AV block and hypotension; severe bradycardia; claudication and cold extremities, Raynaud's phenomenon; dyspnea; palpitations; precordial pain.

CNS: dizziness, lethargy, weakness, drowsiness, headache, insomnia, fatigue, anorexia, anxiety, mental depression, poor concentration, reversible amnesia and catatonia, vivid dreams with or without insomnia, hallucinations, paresthesia, incoordination.

Gastrointestinal: nausea, vomiting, epigastric distress, anorexia, bloating, mild diarrhea, constipation.

Respiratory: bronchospasm, laryngospasm, and respiratory distress (see Contraindications and Warnings).

Dermatologic: A few cases of erythematous rashes and increase of facial acneiform lesions have been reported; urticaria; exfoliative psoriasiform eruption.

Miscellaneous: Reduction or loss of libido; reversible alopecia and, rarely, diminution and loss of hearing; tinnitus; visual disturbances; diminished vision; conjunctivitis; thrombocytopenic purpura; pharyngitis and agranulocytosis, fever combined with aching and sore throat; flushing of the face. Clinical laboratory test findings show elevated blood urea levels in patients with severe heart disease, elevated serum transaminase, alkaline phosphatase, and lactate dehydrogenase.

Overdose:

Symptoms:

Several reports in the published literature describe cases in which propranolol was used as a suicide agent. In most cases, other agents (e.g., alcohol) have also been involved. One patient, thought to have ingested 3600 mg of propranolol died. However, the survival of patients who took higher single doses has also been reported. Common signs to be expected in overdosage are bradycardia, hypotension, bronchospasm, or acute cardiac failure. If overdosage occurs, discontinue therapy with propranolol and observe the patient closely. In addition, the following therapeutic measures are suggested.

Treatment:

Bradycardia: Administer atropine incrementally in 0.6 mg doses. If there is no response to vagal blockade, administer isoproterenol cautiously.

Cardiac failure: digitalization and diuretics.

Hypotension: Vasopressors, e.g., levarterenol or epinephrine (there is evidence that epinephrine is the drug of choice).

Bronchospasm: Administer isoproterenol and aminophylline.

Dosage:

Hypertension: Therapeutic response to a given dosage varies between patients; therefore, dosage must be individually titrated and should be carefully monitored. Therapy may be started by administering the drug in 2 equal daily doses of 40 mg. This may be increased. If necessary, in 1 week, to 80 mg twice daily, before

breakfast and at bedtime. If necessary, the drug may be increased to 160 mg twice daily. For most patients the dosage is within the range of 160 to 320 mg daily. A small number of patients may respond to 80 mg daily. Experience to date suggests that in some resistant patients increasing the dosage above 320 mg/day may have an additional effect. Give doses above 320 mg/day on a 3 or 4 times daily regimen. The time course of full blood pressure response is variable. The antihypertensive effect will usually occur within 3 to 7 days after reaching the effective dose. The maximum decrease in blood pressure may occur 2 to 4 weeks after starting treatment. Angina Pectoris: Individualize dosage. Starting with 10 to 20 mg 3 or 4 times daily, before meals and at bedtime, increase dosage gradually at 3 to 7 day intervals until optimum response is obtained. Although individual patients may respond at any dosage level, the average optimum dosage appears to be 160 mg per day. Occasionally, in resistant patients, dosage as high as 320 to 400 mg per day have been administered with beneficial results. If treatment is to be discontinued, reduce dosage gradually over a period of about 2 weeks (see Warnings).

Arrhythmias: Before meals and at bedtime, 10 to 30 mg 3 or 4 times daily.

Migraine: Individualize dosage. The initial dose is 40 mg twice daily. The dose may then be gradually increased until optimum migraine prophylaxis is achieved. The usual effective dose range is 80 to 160 mg/day.

Hypertrophic Subaortic Stenosis: Before meals and at bedtime, 20 to 40 mg, 3 or 4 times daily.

Pheochromocytoma: The preoperative dosage is 60 mg daily, in divided doses, for 3 days before surgery, concomitantly with an alpha-receptor blocking agent. For malignant cases, 30 mg daily in divided doses.

Mode of Administration: Oral

Availability Of Dosage Form :

Apo-Propranolol 10 mg :

Each round, orange, biconvex tablets, scored and engraved "APO" over "10" on one side, plain on the other contains 10 mg propranolol. Available in bottle of 100 tablets and PVC Blisters of 1000 tablets. Not all packaging is marketed.

Apo-Propranolol 40 mg :

Each round, green, biconvex tablets, scored and engraved "APO" over "40" on one side, plain on the other contains 40 mg propranolol. Available in bottle of 100 tablets and PVC Blisters of 1000 tablets. Not all packaging is marketed.

Storage Conditions :

Store below 30°C.

Manufacturer: Apotex Inc., 150 Signet Drive, Weston, Ontario, Canada M9L 1T9.

Date of Revision of Package Insert: 26 July 2024

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ST 0052C-3